

Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 2nd mid-term test– December 22nd 2022

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:30m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about Parallel Algorithms and Parallel Programming and briefly explain (without an explanation, the answer will be considered invalid)

A. Could you please summarize why manual parallelization is preferred to automatic parallelization? (1)

B. Designing a «good» parallel algorithm by extracting all the available parallelism is **not enough**. True/False? Why?(1)

C. Loop-carried dependences prevent loop parallelization. True/False? Why?(1)

D. Is an output dependence a true dependence? True/False? Why? (1)

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please define what a parallel reduction is: assumptions, complexity, example.(1)

B. Please describe how a parallel pattern gather works. (1)

C. Describe all the collision rules for a scatter pattern. (1)

D. Describe how the pack parallel pattern could be parallelly implemented. (1)

Exercise n. 3

Answer the following questions about pthreads and OpenMP and briefly explain (without an explanation, the answer will be considered invalid)

A. Using Pthreads, is it possible to synchronize the execution flow of multiple threads so that they will progress beyond a given point only after all threads have reached it? (1)

B. What's the use of the omp single directive? (1)

C. How can we control the number of threads spawned by the omp parallel pragma? (2)

Exercise n. 4

A. Describe the functionality of the MPI_Gather subroutine (You can skip this question if you completed the second challenge) (2)

B. How does MPI differ from Pthreads and OpenMP? (2)